Weakly Supervised Sea Fog Detection in Remote Sensing Images via Prototype Learning

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***Abstract*— Sea fog detection is a challenging and significant task in the field of remote sensing. Deep learning-based methods have shown promising potential, but require a large amount of pixel-level labeled data that are time-consuming and labor-intensive to acquire. To scale up the dataset and overcome the limitations of pixel-level annotation, we attempt to explore the existing knowledge from historical statistics for label-efficient sea fog detection. In this article, we propose an image-level weakly supervised sea fog detection dataset (WS-SFDD) and a novel weakly supervised sea fog detection framework via prototype learning, named ProCAM. According to the sea fog events recorded by the Marine Weather Review published quarterly by the National Meteorological Center of China, we collect the sea fog images from Himawari-8 satellite data and obtain free image-level labels to construct the dataset. However, with image-level annotations, the existing weakly supervised semantic segmentation (WSSS) methods mainly rely on class activation maps (CAMs) and have limitations when applied to such a specific scenario: 1) the pseudo-labels (PLs) mainly cover the most discriminative part of object regions that are incomplete; 2) the background is complex with varying atmospheric conditions, and it is difficult to distinguish sea fog from low clouds due to their high similarity in spectral characteristics; and 3) the co-occurring context, such as "sea," distracts the model and thus degrades the performance. To address the above issues, in our proposed ProCAM, we first design a prototype reactivation (PRA) module that reactivates self-similar sea fog regions by pixel-to-prototype feature matching to improve the robustness and completeness of CAMs. Then, we develop a pixel-to-prototype contrastive (PPC) learning method to increase the distance between sea fog and background in the embedding space for learning more discriminative dense features. Finally, a self-augmented regularization (SAR) strategy is presented to decouple sea fog from its co-ocurring context, and thus avoid background interference. Extensive experiments on the WS-SFDD dataset demonstrate that our proposed method ProCAM achieves superior performance with an $F1$ score of 77.59% and a critical success index (CSI) of 63.39%. To the best of our knowledge, this is the first work to perform image-level weakly supervised sea fog detection in remote sensing images. The dataset and code are available at https://github.com/yixianghuang/ProCAM.**

***Index Terms*— Prototype learning, remote sensing image segmentation, sea fog detection, weakly supervised learning.**

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I. Introduction

SEA fog is a kind of disastrous weather that commonly occurs over the sea surface or coastal areas, formed by the condensation of water vapor [1]. In foggy conditions, the horizontal visibility is less than 1 km [2], thus severely affecting maritime transportation, navigation, and safety. Hence, accurate sea fog detection is a highly demanding and significant task.

Conventional sea fog detection relies on meteorological stations or buoys, which are distinctly sparse in spatial and temporal distribution [3]. In recent years, the development of satellite remote sensing technology has facilitated large-scale and long-term sea fog observations. Remote sensing imagery provides a cost-effective and efficient way to monitor sea fog over large areas and can help predict its occurrence and movement.

Based on remote sensing data from satellites, the existing methods can be mainly categorized into two groups.

1) *Traditional Thresholding Techniques:* These techniques depend on handcrafted features and empirical thresholds, e.g., a series of bright temperature difference (BTD) approaches [4], [5], [6]. It is difficult to set appropriate thresholds to deal with different complex scenarios, resulting in the limitations of accuracy and robustness.
2) *Machine Learning Algorithms:* These algorithms, especially deep learning algorithms, have achieved promising performance in the task of sea fog detection. However, a large amount of pixel-level labeled data are required for the fully supervised methods, which can be labor-intensive and time-consuming to obtain. A large-scale pixel-level annotated sea fog detection dataset (SFDD) is proposed in [7], which supports deep learning-based sea fog detection. However, it requires expensive manual labeling with domain expertise to further scale up the dataset.

To relieve the burden of pixel-level data annotation, several studies have made new attempts by introducing transfer learning [8] or weakly supervised methods [9]. Xu et al. [8] transfer the knowledge of land fog to sea fog by unsupervised domain adaptation, requiring manual annotations on source domain. Zhu et al. [9] utilize a handful of manual point annotations for both foreground and background to generate pseudo-labels (PLs) for model optimization, but still relies on the visibility data from sparse meteorological stations and buoys. Also,

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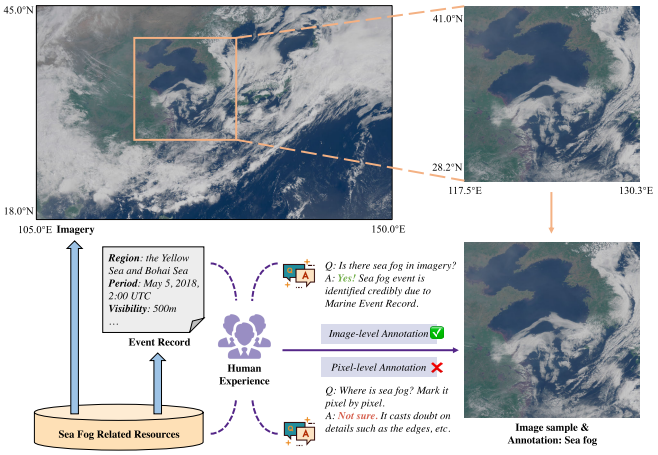


Fig. 1. Illustration of our motivation. We can mine the existing knowledge from freely available historical statistics for label-efficient sea fog detection.

visibility is not a definitive indication of the occurrence of sea fog.

Moreover, it is a hard issue to distinguish sea fog and low clouds due to their high similarity in spectral characteristics, which is also a challenging task for human annotators to recognize exact sea fog regions given a remote sensing image. Relatively, it is much easier to tell if there is sea fog in this image and obtain an image-level label indicating the presence of sea fog. We found that the Marine Weather Review provides a detailed account of the occurrence of sea fog in China's coastal waters over the past few years, including specific regions and periods. Such valid information can be converted into image samples with image-level labels for weakly supervised sea fog detection. This prompted us to think about how to make full use of weak supervision to train a robust sea fog detection model.

In this article, our motivation is to mine the existing knowledge from freely available historical statistics for label-efficient sea fog detection, as shown in Fig. 1. To this end, we build a weakly supervised SFDD (WS-SFDD) upon remote sensing images and explore sea fog detection from a new perspective of image-level weak supervision. Specifically, we crop and label the images by region and period according to the Marine Weather Review, as detailed in Section III. It is inexpensive to scale up the dataset without the limitations of pixel-level manual labeling.

With image-level annotations, we attempt to address the sea fog detection task by introducing weakly supervised semantic segmentation (WSSS) technology. Most WSSS approaches are based on class activation maps (CAMs) [10] and follow a multistage learning procedure: producing an initial CAM, generating pseudo-masks, and training a segmentation model. Nevertheless, not all WSSS approaches for natural scene images can be appropriately applied to such a binary segmentation task in remote sensing images. In such a scenario, there remain some challenges and issues to be addressed, as shown in Fig. 2.

1) Incomplete Activation: Image-level labels inherently lack spatially detailed cues and object structure information, which are crucial for dense prediction tasks in remote sensing images with complex scenes. CAM tends to activate only the most discriminative part of the object regions, resulting in false negative (FN) PLs that degrade the model performance. Meanwhile, sea fog may be located anywhere in the image, and its region is irregular, so it is difficult to extract the most significant region correctly. To obtain more integrate CAMs, we propose a weakly supervised sea fog detection framework via prototype learning, named ProCAM. Considering the self-similarity of sea fog, a prototype can well represent the distribution of sea fog features. This inspires us to exploit the initial CAMs to generate a class-specific prototype to match highly similar dense features that may not have been activated.

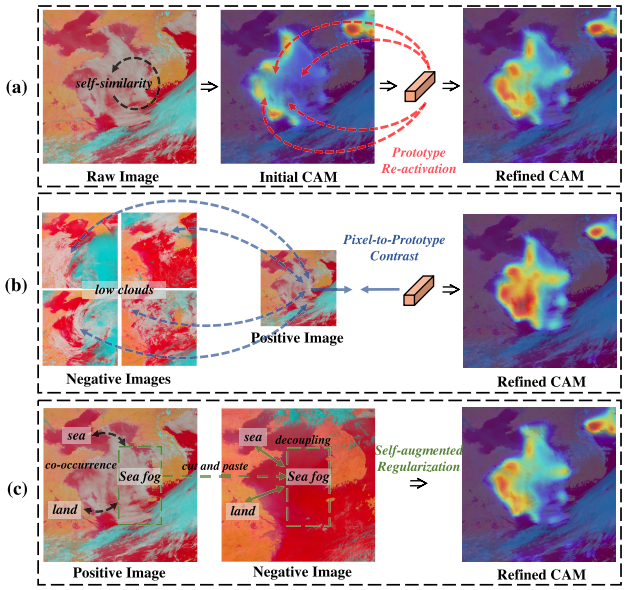


Fig. 2. Challenges and issues in weakly supervised sea fog detection. (a) Incomplete activation. (b) Complex background. (c) Coupled context.

2) Complex Background: In sea fog images, the background is complex with varying atmospheric conditions. For the sea fog detection task, it is a tough problem to distinguish sea fog from low clouds, especially in mixed cloud and fog areas, which drives us to devise a pixel-to-prototype contrastive (PPC) learning method. The objective is to increase the interclass distance between sea fog and background objects in the embedding space, making sea fog more distinguishable. However, in such a complex scene with diverse categories, it is not desirable to describe the class distribution of background with a single prototype. Instead, we take the regional semantic features, i.e., dense background features, from negative images as the negatives for contrastive learning, which realizes cross-image pixel contrast.

3) Coupled Context: In such a remote sensing scene, sea fog is always present with its co-occurring context, such as "sea," which leads the model to pay less attention to the object features. Therefore, we present a self-augmented regularization (SAR) strategy to disassemble the inherent contextual information. Specifically, we crop the sea fog regions out of the positive image with the initial CAM mask and paste them into the negative image to obtain a mixed image. Then, we use the PLs to supervise the CAM predictions of the augmented image with an $L2$ loss.

In summary, our main contributions are as follows.

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1) We construct an image-level annotated WS-SFDD and propose a weakly supervised sea fog detection framework via prototype learning (ProCAM) as the first benchmark for image-level weakly supervised sea fog detection in remote sensing images.
2) A prototype reactivation (PRA) module is proposed to reactivate self-similar sea fog regions by pixel-to-prototype feature matching, thus enhancing the robustness and completeness of CAMs.
3) A PPC learning method is devised to regularize the embedding space by explicitly enforcing intraclass compactness and interclass dispersion, thus learning more discriminative dense features.
4) An SAR strategy is presented to decouple sea fog and its co-occurring context by region-aware cut and paste, driving the model to focus more on foreground object features.

II. Related Work

A. Sea Fog Detection

With the advancement of satellite remote sensing technology, the work related to sea fog detection based on remote sensing imagery has been extensively studied in recent years. Various methods have been proposed to address the challenges of sea fog detection, including traditional thresholding techniques and machine learning algorithms.

Traditional thresholding techniques are the simplest and most widely used methods for sea fog detection in remote sensing imagery. These techniques involve setting a threshold value for the pixel intensity of the image and classifying pixels that exceed this threshold as sea fog. Zhang et al. [11] determined the sea fog monitoring threshold and established a daytime sea fog detection model based on the analysis of visible (VIS) and infrared radiation characteristics of different underlying surfaces, such as ocean, mid-high cloud, low cloud, and sea fog. Wu and Li [5] proposed a dynamic threshold scheme and designed an automatic sea fog detection method based on dual-channel BTD. Shang et al. [12] developed a daytime cloud and haze detection algorithm by incorporating altitude information to adjust the thresholds. However, these methods have limitations in terms of accuracy and reliability, especially in the presence of clouds and other atmospheric conditions.

Machine learning algorithms, especially deep learning algorithms, have shown promising potential in this task. These algorithms involve training a deep neural network on a large dataset of labeled images to learn the features and patterns of sea fog. Chunyang et al. [13] trained a U-Net model with Moderate-Resolution Imaging Spectroradiometer (MODIS) multispectral images and evaluated its effectiveness in sea fog detection. Huang et al. [14] proposed a correlation context-driven sea fog detection method by exploiting pixel–superpixel relationships and pixelwise dependencies to achieve more refined segmentation results (SRs). Zhou et al. [7] proposed a per-pixel annotated dataset to support deep learning-based sea fog detection and designed a dual branch neural network to extract discriminative features from both visual and statistical domains. However, these supervised learning methods rely on a large amount of pixel-level labeled data, which can be expensive and time-consuming to obtain.

New attempts have been made to solve the label-scarcity problem by introducing transfer learning or weakly supervised learning approaches. Xu et al. [8] proposed an unsupervised domain adaptation method to transfer the knowledge learned from land data to the sea fog detection task, which still requires manual annotations on source domain. Zhu et al. [9] proposed a point-based weakly supervised sea fog detection method with the auxiliary information of International Comprehensive Ocean-Atmosphere Data Set (ICOADS) visibility data, but rely on fixed and limited observations and buoys at sea that are not reliable.

In contrast to the above methods, we do not require any pixel-level manual annotation and make full use of historical statistics to construct an image-level labeled sea fog dataset. Based on this, we propose a novel weakly supervised method via prototype learning to achieve label-efficient deep sea fog detection. To the best of our knowledge, we are the first to propose a weakly supervised sea fog detection method with image-level annotations only.

B. Weakly Supervised Semantic Segmentation

To relieve the burden of pixel-level labeling, numerous WSSS methods have been proposed. According to different labeling types, these methods can be divided into four categories: scribbles [15], points [16], bounding boxes [17], [18], [19], and image labels (WSSS-IL) [20], [21], [22]. Our work concentrates on how to utilize image-level annotations to execute the sea fog detection task in remote sensing images as the most economical among the aforementioned weak supervisions.

Most WSSS-IL methods rely on CAMs [10], [23], [24] of the classification model to obtain the seed areas that cover discriminative semantic regions in the image. As shown in Fig. 3, the dominant approaches for WSSS-IL follow a multistage learning procedure: producing an initial CAM, generating pseudo-masks, and training a segmentation model [25]. Most studies focus on improving the quality of the initial seed (e.g., CAM) to generate more reliable pseudo-masks, such as region growing [26], erasing [27], [28], [29], [30], affinity learning [31], [32], [33], and decoupling [34], [35]. IRNet [20] propagated the seed areas to reach the boundary and utilized the interpixel relationship to refine the final pseudo-masks. AdvCAM [21] proposed an antiadversarial framework to manipulate the CAMs by perturbing images along pixel gradients related to the classification of the target class. ReCAM [22] proposed to reactivate the converged CAM with binary cross-entropy (BCE) loss by using softmax cross-entropy (SCE) loss. Some works also exploit additional prior information for optimization, such as saliency map [36], [37], cross-image relation [38], [39], [40], and out-of-distribution data [41]. Our work follows the mainstream pipeline and focuses on enhancing the robustness and completeness of CAMs.

Weakly supervised sea fog detection is a specific application of WSSS on remote sensing images. The aforementioned approaches mostly focused on natural scene images, but rarely explored binary segmentation to extract foreground object in

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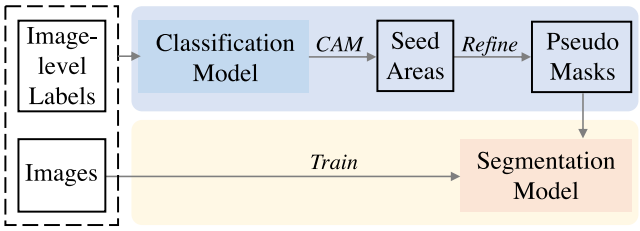


Fig. 3. Mainstream pipeline for WSSS with image-level labels.

remote sensing images [42], such as road [43], [44], building [45], [46], [47], and water [48]. To the best of our knowledge, we are the first to achieve sea fog detection under image-level weak supervision.

C. Prototype-Based Weakly Supervised Learning

Prototype-based weakly supervised learning is a specific area of research that uses prototypes to represent the underlying class distribution, which can help to overcome the challenges of weak supervision. Du et al. [49] proposed prototype-based contrastive learning within a view and across different views of an image to learn a more compact feature space. Differently in contrastive objectives, our work performs cross-image pixel contrast with dense background features as negatives rather than a single prototype that cannot well represent the class distribution of diverse backgrounds. Zhou et al. [50] proposed regional semantic contrast and aggregation, which gathers diverse relational contexts in the memory and performs dataset-level relation learning. However, the pseudo-region embeddings in the memory bank will introduce unavoidable noise, and our work randomly samples background embeddings from negative images instead. Chen et al. [51] proposed self-supervised image-specific prototype exploration that tailors prototypes for every image to capture complete regions and imposes consistency regularization between original CAM and image-specific CAM. It evaluates the classwise structure similarity by intersection over union (IoU), while our work calculates the cosine similarity to obtain class-specific activation mask by threshold. In contrast to these approaches, our work considers the characteristics of sea fog images and designs a specific method for the sea fog detection task.

III. Weakly Supervised Sea Fog Detection Dataset

Deep learning-based sea fog detection requires a large amount of labeled data, with high demands on both sample diversity and label accuracy. Most previous work relied on station-level annotations from sparse observations and buoys at sea. In [7], a pixel-level annotated SFDD is proposed, but it requires laborious manual labeling and specialized domain knowledge. Moreover, it is extremely expensive to scale up the dataset. We discovered that free image-level labels are available from the existing historical statistics that reported accurate but inexact sea fog occurrences and, therefore, propose to address the sea fog detection task from a novel perspective of image-level weak supervision. To the best of our knowledge, the proposed WS-SFDD is the first WS-SFDD with large-scale quantity, long-time span, and image-level labeling.

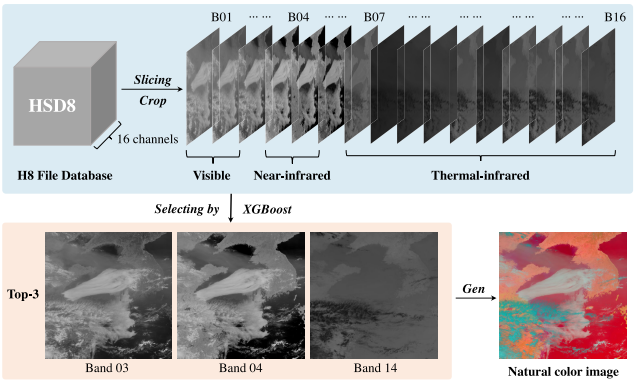


Fig. 4. Details of HSD8.

A. Data Description

The satellite data used to construct the dataset are the multispectral Himawari-8 Standard Data (HSD8) from Himawari-8. Himawari-8 is a geostationary meteorological satellite operated by the Japan Meteorological Agency to perform meteorological and environmental observation missions [52]. Compared with other satellites, the HSD8 reaches 16 channels in three VIS, three near-infrared (NIR), and ten thermal-infrared (TIR) bands with higher temporal and spatial resolution. The spatial resolution of Himawari-8 reaches 0.5–1 km in VIS and 1–2 km in NIR/TIR bands. Himawari-8 provides satellite images at 10-min intervals, allowing high-frequency observations of sea fog cover changes in the regions of interest. Following the previous practice [8], [9], [14], we evaluated the 16 channels by extreme gradient boosting (XGBoost) [53] to select the top-3 significant ones (3, 4, and 14) for sea fog in discriminative features to generate natural color images, as shown in Fig. 4. This effectively discards redundant information and significantly reduces the computational complexity. We used these natural color images as experimental data instead of the full 16 channels.

B. Dataset Construction

The Marine Weather Review issued by the National Meteorological Center of China has recorded sea fog events by season from 2017 to 2021, providing detailed descriptions of specific dates and areas of sea fog. We selected the Yellow Sea and the Bohai Sea as our study area, where sea fog incidents occur frequently. The region of interest (28.2°N–41.0°N, 117.5°E–130.3°E) is cropped from the whole satellite image. From the Marine Weather Review, it is naturally easy to obtain accurate image-level labels without the need for humans to distinguish the form of sea fog. For each image, 0 means nonfog (negative sample), and 1 means sea fog (positive sample). According to historical statistics, we collected, filtered and sorted 4291 satellite images with a size of 1024×1024 from 2017 to 2021 to construct the WS-SFDD. The collected sea fog images were manually selected by meteorological experts to ensure the presence of sea fog in the

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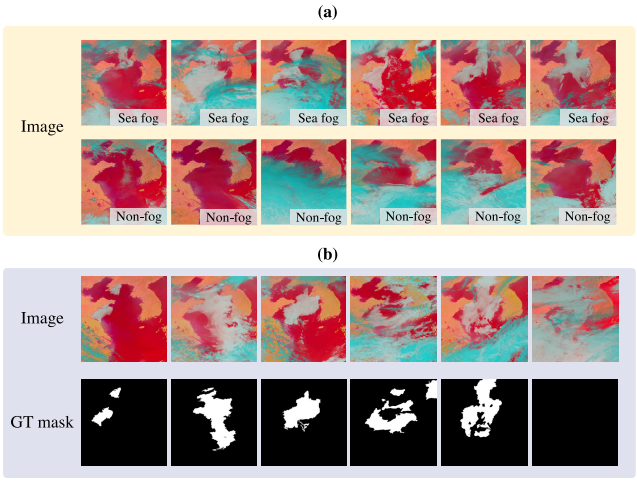


Fig. 5. Details of (a) training set and (b) test set in the WS-SFDD.

image if a sea fog event was recorded. In the case of cloud cover, partial coverage was retained, and almost complete cover (about 80%) was discarded. Specifically, the dataset consists of 3935 image-level annotated images for training and 356 pixel-level annotated images for test, as shown in Fig. 5. The ratio of positive and negative samples is approximately 1:1. For the test set, sea fog areas are manually labeled by meteorological experts from the National Meteorological Center of China, according to the time-series data from Himawari-8 meteorological satellite, the published Marine Weather Review, and the visibility information recorded by the observation stations. Each expert makes a double-checked correction based on their experience.

IV. Method

In this section, we elaborate the proposed ProCAM framework for weakly supervised sea fog detection, as illustrated in Fig. 6. First, we briefly review the preliminary of CAM. Secondly, we describe the pipeline of exploring class-specific prototype to perform PRA and pixel-to-prototype contrast. Then, an SAR strategy is introduced to decouple co-occurring context and avoid background interference. Finally, to further improve the performance, we also employ the pixel-adaptive refinement (PAR) module used in [54] as postprocessing.

A. Revisiting CAMs

CAM [10] aims to train a multilabel classification model with global average pooling (GAP) followed by a fully connected (FC) layer, optimized by a multilabel soft cross-entropy classification loss

$$\mathcal{L}_{\text{cls}} = -\frac{1}{K}\sum_{k=1}^{K} y[k]\log\sigma(p[k]) + (1-y[k])\log(1-\sigma(p[k])) \tag{1}$$

where $p[k]$ denotes the prediction logits of the kth class, $\sigma(\cdot)$ is the sigmoid function, and K is the number of categories. $y[k] \in \{0,1\}$ is the image-level label for the kth class, where 1 denotes the class is present and 0 otherwise.

Given an image $\mathcal{I}$ from the training set, the per-class object activation map is calculated as follows:

$$\text{CAM}_k(\mathcal{I}) = \frac{\text{ReLU}\left(\Theta_k^{\top}\mathcal{F}\right)}{\max\left(\text{ReLU}\left(\Theta_k^{\top}\mathcal{F}\right)\right)} \tag{2}$$

where Θ_k is the classification weights for the kth class, $\mathcal{F} = f(\mathcal{I})$ denotes the feature maps of $\mathcal{I}$ before GAP, and max indicates taking the maximum element value of the vector.

B. Prototype Reactivation

ProCAM starts with the results of CAM model, which provides the initial CAMs as in (2). For CAM, the prediction logits can be denoted as follows:

$$p = \text{FC}_1(\text{GAP}(f(\mathcal{I}))). \tag{3}$$

To mine the cross-image contexts, we propose a label-aware training paradigm that loads paired positive and negative samples for classifier training. The input positive and negative images $\{\mathcal{I}_{\text{pos}}, \mathcal{I}_{\text{neg}}\}$ are processed by a weight-shared backbone to extract image features $\{\mathcal{F}_{\text{pos}}, \mathcal{F}_{\text{neg}}\} \in \mathbb{R}^{H\times W\times D}$, where $H \times W$ is the spatial size and D is the feature dimension. We extract the class-specific CAM_{pos} for sea fog from the positive image $\mathcal{I}_{\text{pos}}$; then, the initial pseudo-mask can be obtained by taking a threshold value β

$$\mathcal{M}_{\text{pos}} = \begin{cases} 1 & \text{if } \text{CAM}_{\text{pos}} > \beta \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

Then, the positive feature $\mathcal{F}_{\text{pos}}$ and its initial pseudo-mask $\mathcal{M}_{\text{pos}}$ are fed into a masked average pooling (MAP) [55] layer to generate the local prototype vector $\mathcal{P}_{\text{local}} \in \mathbb{R}^D$ for foreground regions, i.e., sea fog

$$\mathcal{P}_{\text{local}} = \text{MAP}\left(\mathcal{M}_{\text{pos}}, \mathcal{F}_{\text{pos}}\right). \tag{5}$$

Inspired by Fan et al. [56], we use the local prototype $\mathcal{P}_{\text{local}}$ as query to match the self-similar features and, thus, reactivate more integrate sea fog regions. We obtain the reactivation mask $\widetilde{\mathcal{M}}_{\text{pos}} \in \mathbb{R}^{H\times W}$ by calculating the cosine similarity between $\mathcal{P}_{\text{local}}$ and $\mathcal{F}_{\text{pos}}$ and taking a threshold δ

$$\widetilde{\mathcal{M}}_{\text{pos}} = \begin{cases} 1 & \text{if cosine}\left(\mathcal{P}_{\text{local}}, \mathcal{F}_{\text{pos}}\right) > \delta \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

Finally, the positive feature $\mathcal{F}_{\text{pos}}$ and its reactivation mask $\widetilde{\mathcal{M}}_{\text{pos}}$ are fed into MAP to obtain class-specific feature for classification. Similar to (1), the PRA loss is defined as follows:

$$\mathcal{L}_{\text{pra}} = -\frac{1}{K}\sum_{k=1}^{K} y[k]\log\sigma\left(p'[k]\right) + (1-y[k])\log\left(1-\sigma\left(p'[k]\right)\right). \tag{7}$$

For ProCAM, the prediction logits for the positive image can be denoted as follows:

$$p'[1] = \text{FC}_2\left(\text{MAP}\left(\widetilde{\mathcal{M}}_{\text{pos}}, \mathcal{F}_{\text{pos}}\right)\right) \tag{8}$$

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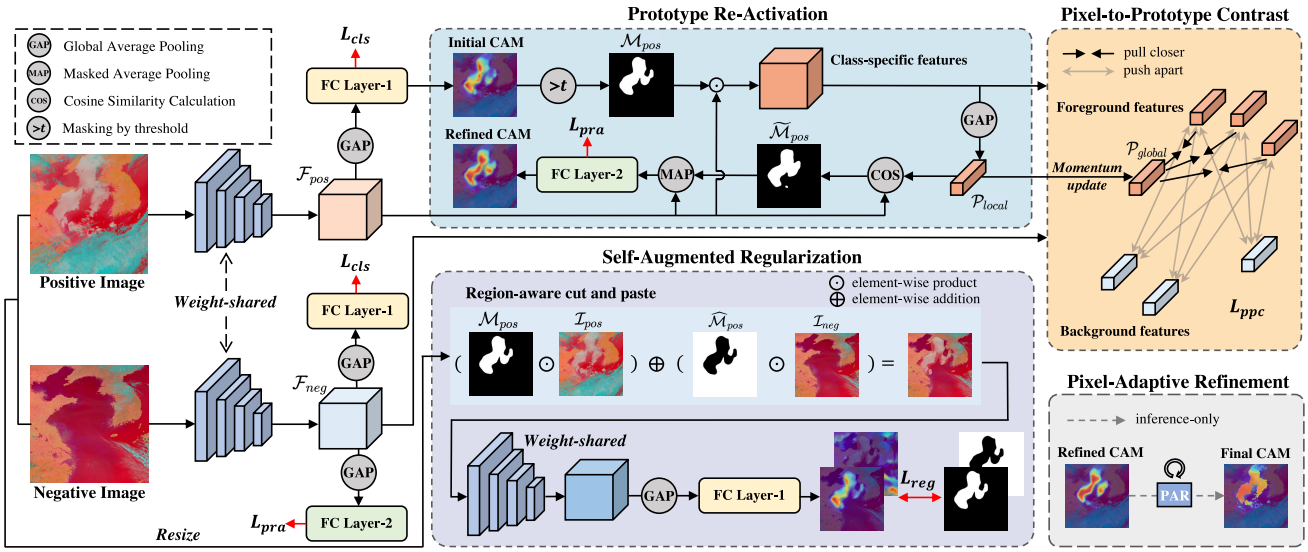


Fig. 6. Detailed architecture of the proposed ProCAM. The framework consists of four components: PRA, PPC, SAR, and PAR. Zoom in for details.

while the prediction logits for the negative image can be denoted as follows:

$$p'[0] = \mathrm{FC}_2\big(\mathrm{GAP}\big(\mathcal{F}_{\mathrm{neg}}\big)\big). \tag{9}$$

Same as in (2), we extract ProCAM as the refined CAM

$$\mathrm{ProCAM}_k(\mathcal{I}) = \frac{\mathrm{ReLU}\big(\Theta_k'^{\top}\mathcal{F}\big)}{\max\big(\mathrm{ReLU}\big(\Theta_k'^{\top}\mathcal{F}\big)\big)}. \tag{10}$$

C. Pixel-to-Prototype Contrast

Due to the high similarity of spectral characteristics between sea fog and low clouds, it is a hard issue to distinguish these two atmospheric conditions. Hence, we perform cross-image semantic contrast over pseudo-pixel embeddings for learning more discriminative dense features. Specifically, we develop a PPC learning method to regularize the embedding space by explicitly enforcing intraclass compactness and interclass dispersion.

With the above initial pseudo-mask $\mathcal{M}_{\mathrm{pos}}$, we can obtain a number of foreground embeddings $\mathcal{N}_f$ from the positive feature $\mathcal{F}_{\mathrm{pos}}$ as the query samples υ. To explore cross-image semantics, we build a nonparametric and dynamic memory bank to store a dataset-level global prototype $\mathcal{P}_{\mathrm{global}}$, which is updated by

$$\mathcal{P}_{\mathrm{global}} \leftarrow \gamma\mathcal{P}_{\mathrm{global}} + (1-\gamma)\mathcal{P}_{\mathrm{local}} \tag{11}$$

where γ is the momentum for memory evolution. The global prototype $\mathcal{P}_{\mathrm{global}}$ is used as the positive sample υ^+. Due to the wide variety and variation of background categories, it is impractical to describe the background distribution with a single prototype. Considering the consistent background elements across images, we extract the background embeddings $\mathcal{N}_b = \{\mathcal{N}_1, \mathcal{N}_2, \cdots, \mathcal{N}_{H\times W}\} \in \mathbb{R}^D$ from the negative feature $\mathcal{F}_{\mathrm{neg}} \in \mathbb{R}^{H\times W\times D}$ as the negative samples υ^- instead. The PPC loss is defined as follows:

$$\mathcal{L}_{\mathrm{ppc}} = \frac{1}{|\mathcal{N}_f|}\sum_{\upsilon\in\mathcal{N}_f} -\log\frac{e^{\mathrm{sim}(\upsilon\cdot\upsilon^+)/\tau}}{e^{\mathrm{sim}(\upsilon\cdot\upsilon^+)/\tau} + \sum_{\upsilon^-\in\mathcal{N}_b} e^{\mathrm{sim}(\upsilon\cdot\upsilon^-)/\tau}} \tag{12}$$

where τ is a temperature coefficient and $\mathrm{sim}(i,j) = (i\cdot j/||i||_2||j||_2)$ is the dot product between ℓ_2-normalized i and j, i.e., cosine similarity. The objective is to make the foreground embeddings closer to the global prototype and away from the background.

D. Self-Augmented Regularization

Inspired by Su et al. [35], we propose an SAR strategy for CAM-based model in a region-aware cut-and-paste manner. In contrast, our method performs online pseudo-mask generation and trains the network end-to-end with a regularization on CAM prediction. Specifically, we crop the sea fog regions out of the positive image $\mathcal{I}_{\mathrm{pos}}$ with the initial pseudo-mask $\mathcal{M}_{\mathrm{pos}}$ and then paste them into the negative image $\mathcal{I}_{\mathrm{neg}}$ to obtain the new mixed image $\mathcal{I}_{\mathrm{mix}}$

$$\mathcal{I}_{\mathrm{mix}} = \big(\mathcal{I}_{\mathrm{pos}}\odot\mathcal{M}_{\mathrm{pos}}\big)\oplus\Big(\mathcal{I}_{\mathrm{neg}}\odot\widehat{\mathcal{M}}_{\mathrm{pos}}\Big) \tag{13}$$

where $\odot$ denotes the elementwise product, $\oplus$ denotes the elementwise addition, and $\widehat{\mathcal{M}}_{\mathrm{pos}} = 1 - \mathcal{M}_{\mathrm{pos}}$. As in (2), we generate $\mathrm{CAM}'_k(\mathcal{I}_{\mathrm{mix}}) = [\mathrm{CAM}'_0, \mathrm{CAM}'_1]$. Correspondingly, we take the pseudo-masks as ground-truth labels $\mathrm{GT}_k(\mathcal{I}_{\mathrm{mix}}) = [\widehat{\mathcal{M}}_{\mathrm{pos}}, \mathcal{M}_{\mathrm{pos}}]$ for supervision. The SAR loss is defined as follows:

$$\mathcal{L}_{\mathrm{reg}} = \sum_{i=1}^{HW}\big(\mathrm{CAM}'_k - \mathrm{GT}_k\big)^2. \tag{14}$$

E. Loss Function

As shown in Fig. 6, our framework consists of four loss terms, i.e., a multilabel soft cross-entropy classification loss

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TABLE I
COMPARISONS OF OUR METHOD WITH OTHER ADVANCED WSSS APPROACHES ON WS-SFDD

Method	CSI		mIoU		Precision		Recall		F1-score	
	PL	*SR*	*PL*	*SR*	*PL*	*SR*	*PL*	*SR*	*PL*	*SR*
CAM [10]	43.92	45.87	69.97	71.07	59.16	61.93	63.03	63.89	61.03	62.89
ReCAM [22]	45.03	47.24	70.49	71.60	57.99	57.47	66.83	72.62	62.09	64.16
AdvCAM [21]	52.89	53.43	74.56	75.22	58.48	70.11	84.70	69.19	69.19	69.64
IRNet [20]	50.26	54.57	73.65	75.99	**74.52**	**80.67**	60.69	62.78	66.89	70.61
ProCAM*	53.33	57.87	75.24	77.59	73.50	72.21	66.02	74.45	69.56	73.32
ProCAM	**63.76**	**63.39**	**80.68**	**80.49**	71.97	72.59	**84.82**	**83.34**	**77.87**	**77.59**

$\mathcal{L}_{\text{cls}}$, a PRA loss $\mathcal{L}_{\text{pra}}$, a PPC loss $\mathcal{L}_{\text{ppc}}$, and an SAR loss $\mathcal{L}_{\text{reg}}$. Formally, the overall loss function is formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{pra}} + \lambda_2 \mathcal{L}_{\text{ppc}} + \lambda_3 \mathcal{L}_{\text{reg}} \tag{15}$$

where λ_1–λ_3 balance the contributions of each loss.

F. Pixel-Adaptive Refinement

Instead of conditional random fields (CRFs) [57] with low inference speed, we exploit an efficient PAR module [54] to postprocess the final pseudo-masks, which incorporates the red, green, blue (RGB) and position of local pixels for label refinement. With low-level image appearance and position information, PAR effectively suppresses the falsely activated regions and enforces better alignment with the low-level boundaries.

V. EXPERIMENT

A. Experimental Setups

1) Dataset and Models: We conduct experiments on the proposed dataset, WS-SFDD. For pseudo-mask generation, we use ResNet-50 [58] as backbone and initialize the parameters with ImageNet-1k [59] pretrained weights. For semantic segmentation, we train the U-Net [60] model with PLs to predict masks for the images in the `test` set. For fair comparison, all the methods use the same backbone network ResNet-50 [58] and segmentation model U-Net [60].

2) Implementation Details: Following a multistage pipeline, we first train an initial CAM model and then train our proposed ProCAM with a batch size of 16 for 10/20 epochs, respectively. We adopt a stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.01, which decays every iteration with a polynomial scheduler [61]. The weight decay factor is set as $1e-4$. For data augmentation, we employ random horizontally flipping and cropping with a cropping size of 512×512. For the U-Net [60] in the step of semantic segmentation, we use dice loss and bootstrapping loss to train the network with a batch size of 32 for 80 epochs. We adopt an Adam [62] optimizer with a learning rate of $1e-4$ and a weight decay of $1e-4$. For efficiency, all the input images are resized to 512×512 during training. We apply test time augmentation (TTA) during inference.

3) Hyperparameter Settings: The default hyperparameters are set as follows. For PRA, the masking/activation threshold (β and δ) is empirically set as (0.2, 0.9). It is experimentally proved that the model is insensitive to the activation threshold δ, and the optimal value is 0.9. For PPC, the temperature τ is 0.1, and the momentum γ is 0.9, following a common practice. For loss function in (15), we set the weight factors (λ_1–λ_3) as (1.0, 0.1, 1.0) to balance the scale of each loss item. For PAR, the dilation rates for extracting neighbor pixels are [1, 4, 16, 32, 48, 64], and the number of iterations is 20, considering the trade-off between accuracy and efficiency.

B. Evaluation Metrics

To provide a quantitative evaluation of our method, we adopt the most standard and convincing metrics: critical success index (CSI), i.e., sea fog IoU, mean mIoU, precision, recall, and $F1$ score. The confusion matrix of sea fog detection results consists of true positive (TP), FN, false positive (FP), and true negative (TN). The specific meanings are as follows.

1) *TP:* Sea fog is existent and detected.
2) *FN:* Sea fog is existent but undetected.
3) *FP:* Sea fog is inexistent but detected.
4) *TN:* Sea fog is inexistent and undetected.

Formally, the above evaluation metrics are defined as follows:

$$\text{CSI} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{16}$$

$$\text{mIoU} = \frac{\text{CSI} + \text{background IoU}}{2} \tag{17}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{20}$$

where background IoU = (TN/TN + FP + FN).

The CSI, precision, and recall measure the model's performance in detecting sea fog, while the mIoU and $F1$ score provide a comprehensive evaluation of the model's detection capability.

C. Comparison With Advanced WSSS Methods

In this experiment, we apply the typical and advanced WSSS approaches to the WS-SFDD dataset for comparison with our proposed ProCAM, including CAM [10], ReCAM [22], AdvCAM [21], and IRNet [20]. Due to the lack of pixel-level labels in the `train` set, we report the results of the `test` set instead.

1) Quantitative Analysis: Detailed quantitative results for CSI, mIoU, precision, recall, and $F1$ score (in %) are presented in Table I. PL denotes pseudo-label, and SR denotes segmentation result. * indicates without using the PAR module. The highest scores in all methods are in bold, and the

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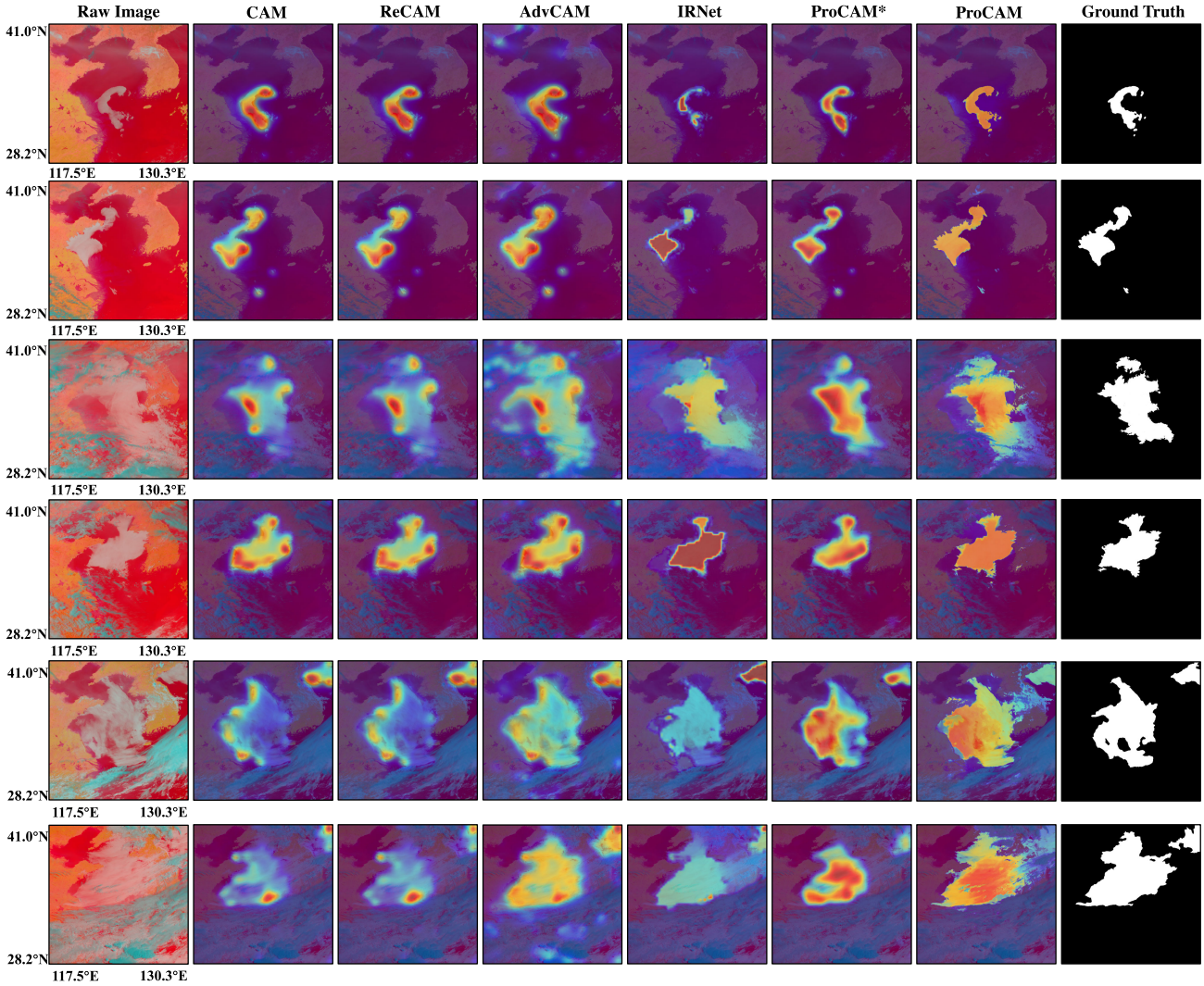


Fig. 7. Visualization of CAMs for sea fog on the WS-SFDD.

TABLE II
ABLATION STUDIES ON DIFFERENT MODULES OF OUR METHOD

Baseline	PRA	PPC	SAR	PAR	CSI	mIoU	Precision	Recall	F1-score
✓					43.92	69.97	59.16	63.03	61.03
✓	✓				44.94+1.20	70.49+0.52	58.92-0.24	65.44+2.41	62.01+0.98
✓	✓	✓			50.10+6.18	73.50+3.53	70.99+11.83	63.00-0.03	66.75+5.72
✓	✓		✓		49.05+5.13	72.70+2.73	61.50+2.34	70.79+7.76	65.82+4.79
✓	✓	✓	✓		53.33+9.41	75.24+5.27	**73.50+14.34**	66.02+2.99	69.56+8.53
✓				✓	55.00+11.08	75.84+5.87	62.85+3.69	81.50+18.47	70.97+9.94
✓	✓	✓	✓	✓	**63.76+19.84**	**80.68+10.71**	71.97+12.81	**84.82+21.79**	**77.87+16.84**

next highest scores are underlined. For a fair comparison, all the methods are based on the same initial CAMs.

It can be observed that ProCAM achieves superior performance to the other methods in terms of all the metrics, except for precision. ProCAM is the only method with CSI over 60%. The comprehensive evaluation metrics mIoU and $F1$ score of ProCAM also remarkably outperform those of the other methods. Among all the methods, ProCAM achieves the most significant improvement over the baseline CAM, which demonstrates the effectiveness and superiority of our approach. Even without the PAR module, ProCAM* still achieves better overall performance compared with other methods.

2) Qualitative Analysis: The qualitative comparison results for CAMs and SRs are shown in Figs. 7 and 8, respectively. Compared with other methods, ProCAM achieves more accurate CAMs and more refined SRs. Specifically, ProCAM activates more complete sea fog regions and obtains more explicit boundaries. CAM and ReCAM only activate partial sea fog regions, resulting in incomplete SRs. AdvCAM recalls the sea fog area as much as possible, but causes a high FP

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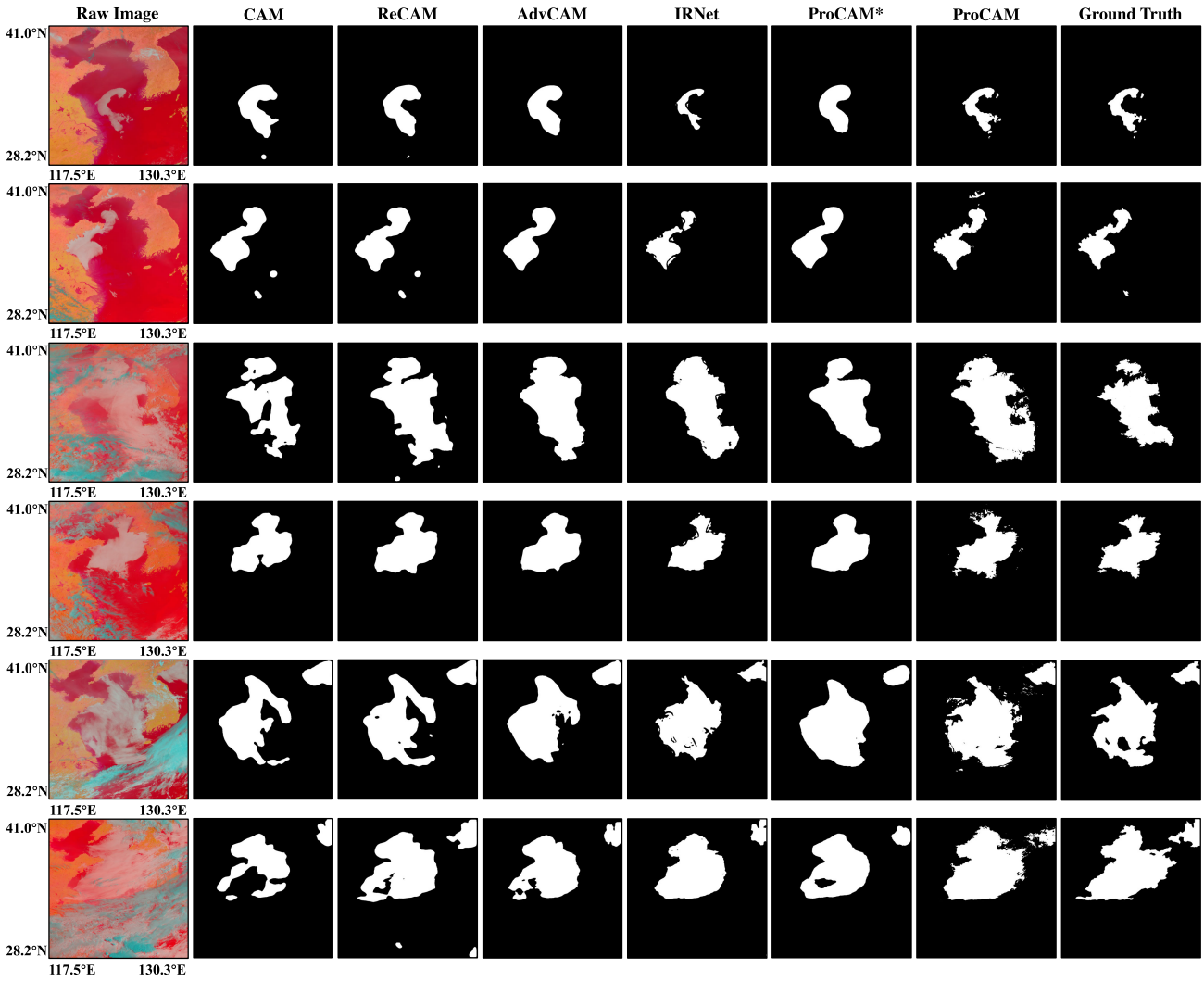


Fig. 8. Visualization of SRs on the WS-SFDD.

rate. IRNet optimizes the object boundaries; however, there still exist undesired FNs. Note that many WSSS methods commonly take IRNet as a refinement option to further optimize the results. Instead, we use the PAR module for postprocessing, which does not require any training. Our proposed ProCAM detects sea fog with high prediction accuracy and low false alarm rate. The final SRs of ProCAM are closer to ground truth.

D. Ablation Study

As seen in Table I, the quality of PLs is critical for training a robust segmentation model. To evaluate the effectiveness of each component in the proposed ProCAM framework, we conduct rigorous ablation experiments in terms of PLs. We utilize CAM as the baseline and systematically equip it with the proposed modules, including PRA, PPC, and SAR. To further refine the predictions, we also test the baseline with the PAR module.

The quantitative results in Table II indicate that each component plays an important part and contributes to a significant improvement over the baseline. The PRA module improves the CSI by 1.20% and, in particular, the recall by 2.41%, which supports our idea of reactivating more integrate sea fog regions by pixel-to-prototype feature matching. On this basis, we add the PPC module and increase the CSI by 4.98%. Especially, the precision is improved by a large margin of 12.07%. This demonstrates that the pixel-to-prototype contrast can help the model learn more discriminative dense features, making sea fog more distinguishable from the background. Instead of PPC, we introduce the SAR module to enhance the feature representation by decoupling sea fog from its co-occurring context, resulting in a 5.35% increase in recall. All the above modules contribute to the best performance of 53.33% CSI, which can even reach 63.76% if we add the PAR module. Our final model still significantly outperforms the baseline with PAR by 8.76%.

The visualization results of CAMs for sea fog are shown in Fig. 9. It can be observed that baseline + PRA activates more self-similar sea fog regions and reinforces the consistency of the activation regions within the target. Baseline + PRA and PPC effectively suppresses redundant activation and reduces

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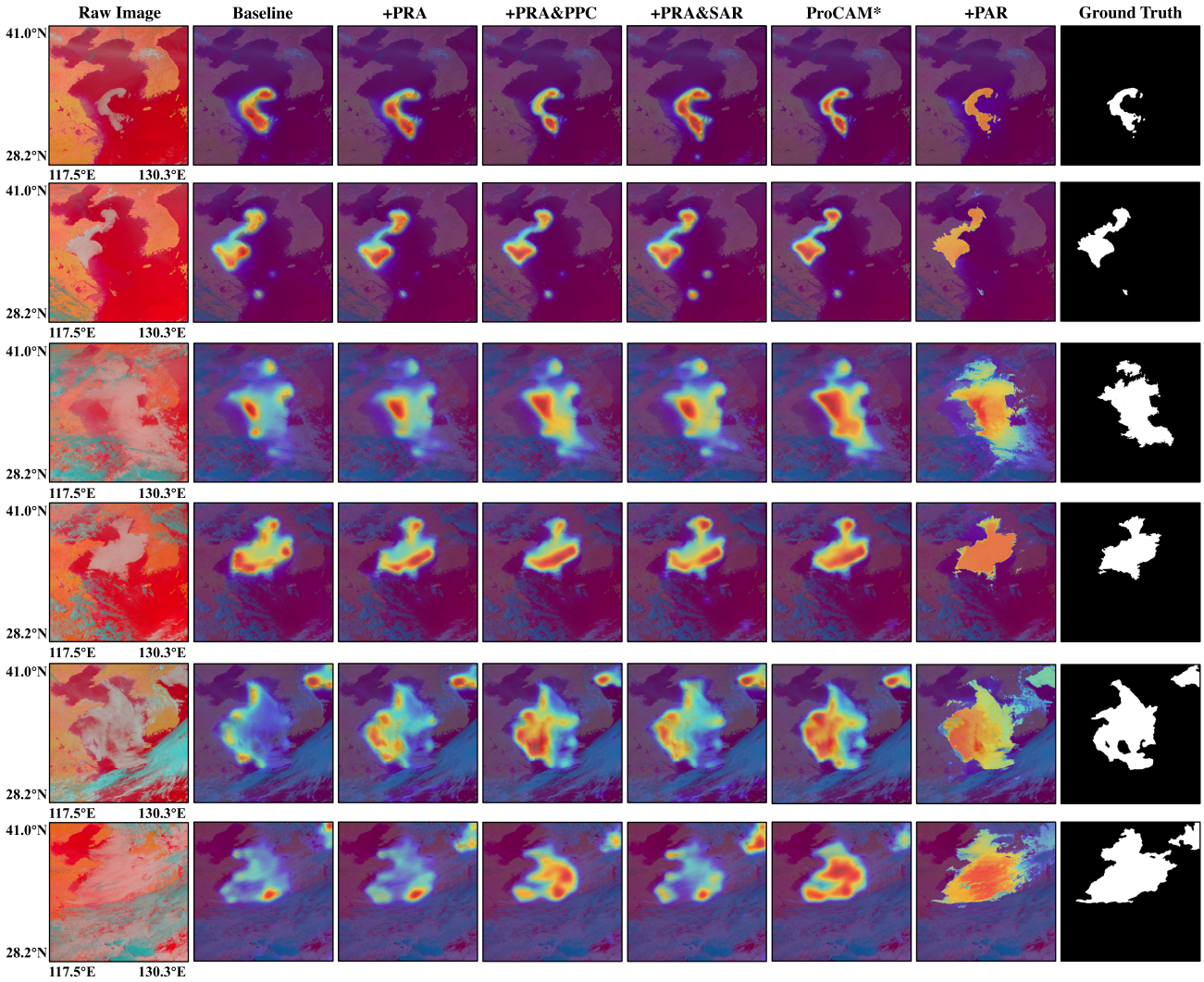


Fig. 9. Visualization of CAMs for sea fog in ablation experiments.

noise, while preserving more complete sea fog regions. Baseline + PRA and SAR improves the localization accuracy especially for small and broken parts. ProCAM* integrates the advantages of all modules to achieve superior overall performance. With the PAR module, ProCAM is capable of producing finer results with sharp boundaries.

To verify the effectiveness of the SAR module, we also visualize the CAMs for background in the negative images, as shown in Fig. 10. To identify the background, most methods except ours only focus on scattered and cluttered regions of the image. Our model pays even attention to almost all regions rather than the most discriminative parts only, attributed to the SAR module. It reveals that the SAR strategy decouples sea fog from the background, which, in turn, enhances the model's focus on the foreground when locating sea fog regions.

E. t-SNE Visualization

To confirm the validity of the PPC module, we perform t-distributed stochastic neighbor embedding (t-SNE) visualization of dense features, as shown in Fig. 11. It can be observed that the background features of CAM are relatively dispersed and grouped into more than one cluster. Some sea fog and background features are mixed together and hard to distinguish. ProCAM yields a more compact embedding space and a more distinct classification boundary. Sea fog and background features are clustered separately.

TABLE III
EFFECTS OF DIFFERENT ACTIVATION THRESHOLD VALUES

δ	CSI	mIoU	Precision	Recall	F1-score
0.7	62.84	80.12	68.66	**88.11**	77.18
0.75	63.16	80.31	69.53	87.33	77.42
0.8	63.55	80.53	70.38	86.76	77.71
0.85	63.72	80.64	70.99	86.15	77.84
0.9	**63.76**	**80.68**	71.97	84.82	**77.87**
0.95	63.62	80.62	**72.68**	83.62	77.77

F. Effects of Activation Threshold

We conduct parametric experiments on the activation threshold δ in (6) to evaluate its effects, as shown in Table III. We can observe that the optimal value of δ is 0.9 in terms of

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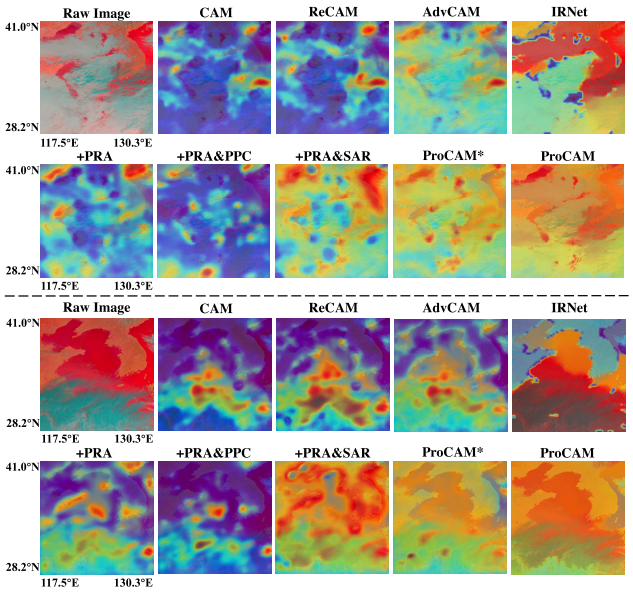


Fig. 10. Visualization of CAMs for background.

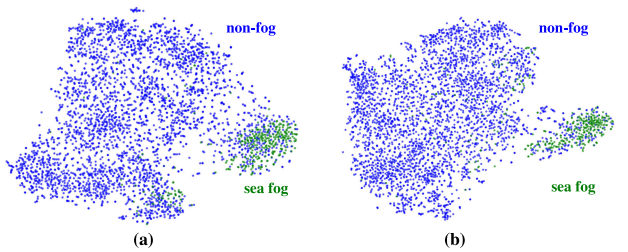


Fig. 11. t-SNE visualization of (a) CAM and (b) ProCAM for dense features on the WS-SFDD.

CSI, mIoU, and $F1$ score, but the difference is not significant when using other values, i.e., ProCAM is not sensitive to δ. The precision rises while the recall declines as the threshold value increases, which demonstrates that δ controls the balance between precision and recall. Lower thresholds introduce more noise, while higher thresholds discard hard positive samples. We set the activation threshold δ to 0.9 by default.

VI. Conclusion

In this article, we propose a novel weakly supervised sea fog detection framework via prototype learning, named ProCAM. It exploits image-level labels to realize pixel-level sea fog detection in remote sensing images. Specifically, ProCAM composes of three critical modules that effectively promote the quality of initial CAMs: PRA, pixel-to-prototype contrast, and SAR. Furthermore, a PAR module is used for postprocessing to obtain more refined PLs and significantly improve the performance of weakly supervised sea fog detection. In future work, our proposed ProCAM approach could be extended to other applications of WSSS beyond sea fog detection, such as cloud detection. By reducing the annotation requirements for segmentation, our approach has the potential to make these applications more accessible and cost-effective. We also plan to further introduce temporal information to improve the accuracy of sea fog detection and utilize multimodal data to address the limitations of observation angles. In addition, we propose a WS-SFDD. To the best of our knowledge, WS-SFDD is the first weakly supervised sea fog detection dataset with large-scale quantity, long-time span, and image-level labeling. It consists of 4291 sea fog remote sensing images collected by Himawari-8, covering sea fog events occurred in the Yellow Sea and the Bohai Sea from 2017 to 2021. Each image is tagged with a credibly accurate label according to freely available historical statistics. This dataset could support further exploration of weakly supervised sea fog detection in remote sensing images.

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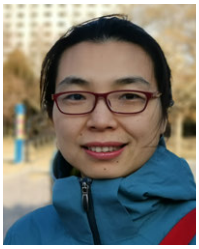
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